Acute MI Hospitalization Adjudication with GPT-5.6 Luna, Gemma 3 4B and 12B, and Qwen3 8B and 32B

Cohort

Stage	Admissions	Patients
Adult admissions (age ≥18)	546,028	223,452
Troponin documented in note or linked labs	59,773	35,370
One admission retained per patient	35,370	35,370
Final analytic sample	250	250

Final analytic sample: 250 unique patients; 75 MI ICD-enriched and 175 non-enriched. The principal-MI ICD proxy contained 44 YES and 206 NO cases.

Eligibility criteria

Domain	Inclusion	Exclusion
Age	18 years or older	Younger than 18 years
Care setting	ICU or ED/emergency/observation admission	No qualifying ICU or ED admission
Discharge summary	Full text; at least 1,500 characters; HPI and hospital-course sections present	Missing, short, partial, stub, truncated, or missing required sections
Troponin	At least one value in the note or linked laboratory data	No documented or linked troponin value
MI ICD enrichment	ICD-9 410.x or ICD-10 I21.x/I22.x used to oversample probable MI	Absence of an MI code did not exclude a case from the non-enriched stratum
Patient deduplication	One hospitalization retained per patient	Repeat admissions excluded after deterministic selection

Exact prompts used

Prompt method	Exact prompt
System instruction	You are a clinical trial endpoint adjudicator. Use only the supplied deidentified hospitalization evidence. Return exactly one word: YES or NO.
Direct	Did this hospitalization involve an acute myocardial infarction? Answer with a single word: YES or NO.
Role	You are a board-certified cardiologist and clinical trial endpoint adjudicator. Determine whether this hospitalization involved an acute myocardial infarction. Answer with a single word: YES or NO.
Structured	Adjudicate acute myocardial infarction using the Fourth Universal Definition. Classify as YES only when BOTH are documented: 1. A rise and/or fall of cardiac troponin with at least one value above the assay's 99th-percentile upper reference limit; AND 2. At least one ischemic feature: ischemic symptoms, new ischemic ECG changes, new pathologic Q waves, new imaging evidence of loss of viable myocardium or a new regional wall-motion abnormality in an ischemic pattern, or intracoronary thrombus. Do not count a remote/history-only MI, chronic stable coronary disease, isolated chronic troponin elevation, myocardial injury without evidence of ischemia, or elective PCI/CABG without periprocedural MI criteria. Answer with a single word: YES or NO.
Fusion	Adjudicate acute myocardial infarction using the Fourth Universal Definition. Use BOTH the discharge summary and the linked structured troponin timeline supplied below. Classify as YES only when there is a rise/fall with at least one troponin above that assay's listed upper reference limit AND at least one ischemic feature in the clinical record. Do not compare values across different assays or units. Do not count myocardial injury without ischemia, chronic elevation, or remote/history-only MI. Answer with a single word: YES or NO.
Input construction	The selected prompt was followed by the discharge-summary evidence wrapper. The linked troponin-timeline wrapper was appended only for Fusion.

Analysis results

Performance was calculated against the principal-MI ICD administrative proxy.

GPT-5.6 Luna

Prompt	TP / TN / FP / FN	Accuracy	Sensitivity	Specificity	PPV	NPV	Kappa
Direct	43 / 179 / 27 / 1	88.8%	97.7%	86.9%	61.4%	99.4%	0.687
Role	43 / 184 / 22 / 1	90.8%	97.7%	89.3%	66.2%	99.5%	0.733
Structured	44 / 177 / 29 / 0	88.4%	100.0%	85.9%	60.3%	100.0 %	0.682
Fusion	36 / 163 / 43 / 8	79.6%	81.8%	79.1%	45.6%	95.3%	0.464

Luna best result: Role prompt => 90.8% accuracy, 97.7% sensitivity, 89.3% specificity, κ = 0.733.

Gemma 3 4B

Prompt	TP / TN / FP / FN	Accuracy	Sensitivity	Specificity	PPV	NPV	Kappa
Direct	43 / 102 / 104 / 1	58.0%	97.7%	49.5%	29.3%	99.0%	0.246
Role	43 / 110 / 96 / 1	61.2%	97.7%	53.4%	30.9%	99.1%	0.277
Structured	31 / 174 / 32 / 13	82.0%	70.5%	84.5%	49.2%	93.0%	0.469
Fusion	43 / 75 / 131 / 1	47.2%	97.7%	36.4%	24.7%	98.7%	0.158

4B best result: Structured prompt => 82.0% accuracy, 70.5% sensitivity, 84.5% specificity, κ = 0.469.

Gemma 3 12B

Prompt	TP / TN / FP / FN	Accuracy	Sensitivity	Specificity	PPV	NPV	Kappa
Direct	32 / 196 / 10 / 12	91.2%	72.7%	95.1%	76.2%	94.2%	0.691
Role	33 / 201 / 5 / 11	93.6%	75.0%	97.6%	86.8%	94.8%	0.767
Structured	17 / 204 / 2 / 27	88.4%	38.6%	99.0%	89.5%	88.3%	0.485
Fusion	38 / 128 / 78 / 6	66.4%	86.4%	62.1%	32.8%	95.5%	0.295

12B best result: Role prompt => 93.6% accuracy, 75.0% sensitivity, 97.6% specificity, κ = 0.767.

Qwen3 8B

Prompt	TP / TN / FP / FN	Accuracy	Sensitivity	Specificity	PPV	NPV	Kappa
Direct	41 / 182 / 24 / 3	89.2%	93.2%	88.3%	63.1%	98.4%	0.686
Role	40 / 190 / 16 / 4	92.0%	90.9%	92.2%	71.4%	97.9%	0.751
Structured	38 / 189 / 17 / 6	90.8%	86.4%	91.7%	69.1%	96.9%	0.711
Fusion	36 / 171 / 35 / 8	82.8%	81.8%	83.0%	50.7%	95.5%	0.522

Qwen3 8B best result: Role prompt => 92.0% accuracy, 90.9% sensitivity, 92.2% specificity, κ = 0.751.

Qwen3 32B

Prompt	TP / TN / FP / FN	Accuracy	Sensitivity	Specificity	PPV	NPV	Kappa
Direct	43 / 184 / 22 / 1	90.8%	97.7%	89.3%	66.2%	99.5%	0.733
Role	43 / 188 / 18 / 1	92.4%	97.7%	91.3%	70.5%	99.5%	0.773
Structured	36 / 192 / 14 / 8	91.2%	81.8%	93.2%	72.0%	96.0%	0.712
Fusion	44 / 123 / 83 / 0	66.8%	100.0%	59.7%	34.6%	100.0 %	0.343

Qwen3 32B best result: Role prompt => 92.4% accuracy, 97.7% sensitivity, 91.3% specificity, κ = 0.773.

Head-to-head model comparison

Model	Top-performing prompt	Accuracy	Sensitivity	Specificity	Kappa;
GPT-5.6 Luna	Role	90.8%	97.7%	89.3%	0.733
Gemma 3 4B	Structured	82.0%	70.5%	84.5%	0.469
Gemma 3 12B	Role	93.6%	75.0%	97.6%	0.767
Qwen3 8B	Role	92.0%	90.9%	92.2%	0.751
Qwen3 32B	Role	92.4%	97.7%	91.3%	0.773